

Myself, My Child or My Friend: Understanding Player Relationships with AI-Generated Game Characters

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Abstract

As AI technology advances in gaming, understanding how players perceive and build relationships with AI-generated characters becomes increasingly important. This study explores players' identity perception patterns of AI characters, focusing on three main relationship types: self-projection (myself), nurturing relationship (my child), and companionship (my friend). Through a two-phase study with 12 initial participants and 120 in the main two-week phase, we examined relationship formation using a custom AI-integrated gaming platform. Data collection combined questionnaires, periodic relationship evaluations, and interaction logs. Our mixed-methods analysis revealed that players' personality traits significantly correlate with their chosen relationship types: introverted players tend to develop nurturing relationships, emphasizing character growth; extroverted players are more likely to establish companionship, valuing social interaction; while younger players often choose self-projection relationships, showing stronger desire for control. These findings provide empirical evidence for designing more personalized gaming experiences with AI characters.

CCS Concepts

• Applied computing → Media arts; • Human-centered computing → Interactive systems and tools.

Keywords

Game Character Generation, Generative AI, Personalized Agents, 2D Character Animation

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1 INTRODUCTION

The identification and recognition of virtual characters constitute fundamental elements of video game experiences [10, 15, 18]. Following Turkle's introduction of "The Second Self" concept [16], which explores how people establish digital identities through technology, this idea has continuously evolved in her subsequent research, suggesting that digital environments allow users to create expressions parallel to their real selves [17]. These characters function not only as visual representations in digital environments but also as virtual entities through which players can establish diverse social relationships [4]. With advancements in artificial intelligence technology, generated game characters are evolving from simple avatars into autonomous agents capable of complex interactions, prompting new investigations into how players perceive and interpret these virtual entities.

Research indicates that players can develop various forms of relationships with game characters, ranging from self-identification to parasocial connections [8]. Over two decades ago, Gee [6] proposed an important theory challenging the user-avatar dichotomy, describing three modes of player-character identity relationships: virtual identity, real identity, and projective identity. Yee and Bailenson's [21] "Proteus Effect" research demonstrates that character representation significantly influences players' actual behavior, further highlighting the necessity of understanding the dynamics between characters and players. Current game research primarily focuses on how players project themselves onto avatars through appearance customization [3] and personality traits [20]. Kafai et al. [9] discovered that players adopt several key strategies when creating characters, including idealized self-creation, new identity



Figure 1: Player-created game characters and their interactions within the game

construction, or disguised identity. Other researchers such as Gazdard [5] proposed the relationship metaphor of "player as parent, character as child," while Boudreau [2] studied how players form hybrid identities in single-player games, developing parent-child-like responsibilities.

The rapid development of artificial intelligence has fundamentally transformed game character creation. AI tools capable of generating diverse 2D images have created unprecedented opportunities for personalized character design, while Large Language Models (LLMs) enable more natural and dynamic character interactions. These technological advancements blur the boundaries between avatars and agents, creating virtual entities capable of meaningful social exchanges. Recent research has explored the impact of automatically generated anime avatars on player identification [19], as well as new paradigms where designers engage with generative AI as co-creative partners in character design [7], reflecting the rapid progress in the field of AI character generation.

However, despite the increasing complexity of AI-generated game characters, there exists a notable gap in systematic research regarding how players construct and perceive relationships with these virtual entities. While existing studies have examined player-character relationships in traditional gaming environments [11, 12], the diversity and complexity of these relationships present new dynamics worthy of investigation in environments with autonomous AI agents, as demonstrated in Sibilla and Mancini's [13] systematic review. Understanding how players categorize and relate to AI-generated characters—whether as extensions of self, virtual dependents, or AI companions—is crucial for designing more engaging and meaningful gaming experiences.

To address this research gap, we have developed a gaming platform integrating generative AI and LLMs to explore player-character relationship patterns. Through experimental studies comparing players' interactions with different types of AI-generated characters, we investigate how players construct and perceive their relationships with these virtual entities. Our research contributions are twofold:

(1) We provide empirical insights into how players categorize and relate to AI-generated game characters, identifying distinct

relationship patterns (myself/my child/my friend) and their implications for game design.

(2) We explore how various aspects of AI character generation and interaction influence players' relationship perceptions, offering design guidelines for creating more engaging and emotionally resonant AI characters.

This paper examines how advances in AI technology are reshaping player-character relationships and identity dynamics in games. By understanding how players perceive and relate to AI-generated characters, we aim to inform the design of more meaningful and socially engaging gaming experiences. Our findings suggest that AI-assisted character creation not only enhances technical capabilities but also opens new possibilities for exploring diverse forms of virtual relationships and social connections.

2 THE GAME

2.1 Design Principles

To evaluate player character identification in AI-assisted game character generation, we considered the following principles when using generative techniques and designing game processes:

- (1) The game can use default character appearances with unbiased LLM driving, or allow players to generate character appearances, attire, race, and personality using natural language. Generated results should fulfill players' imagination without restrictions. Players can also opt for randomly system-prompted characters.
- (2) Game characters should be operable within a scene. Players should be able to view their own and others' characters, freely drive their actions, and interact with other characters, rather than being mere icons.
- (3) In-game, players should be able to converse with other players' characters and NPCs, enhancing character identification through social attributes for a more immersive experience.
- (4) Generated characters' race, occupation, personality, and other attributes should be unbiased at both the model and training set levels, while meeting user requirements.

2.2 Game Platform

Our game is based on an AI interactive character creation framework, Text2AC[14]. Users begin by entering a natural language description of the character’s appearance. They can provide detailed descriptions of the character’s race, clothing style, facial expressions, clothing types, or even eyelash colors, or simply specify gender. After completing the description and clicking "Generate," the system swiftly produces a character illustration(tachie) and a dynamically drivable game character.

Furthermore, users can describe the game character’s personality, background, and occupational skills using natural language. The system seamlessly integrates these inputs into the character panel. Users can then view and examine their created characters. We process users’ natural language descriptions through the Text2AC character generation framework, and convert them into highly flexible character assets using techniques such as image segmentation and automatic skeleton mapping.

Building upon this foundation, we integrated the aforementioned generative framework into the Unity game engine, enabling it to facilitate user interaction and development through natural language under LLM guidance. After creating a game character, users can interact with their character via a text box (Figure 1), directing its actions within the game. Users can also engage in social interactions with characters created by others, such as approaching another character or establishing relationships.

Natural language inputs allow users to command their character to perform a wide range of expressive actions, including jumping, bowing, laughing heartily, expressing anger, or attacking. Character communication logs are recorded as text in a sidebar interface. Additionally, real-time character skeletal animations can be generated through natural language inputs. During dialogue and interaction, the LLM driving the character will comprehensively consider the background and personality of the characters, and generate dialogue and interaction information and animations in real time(Figure 2).

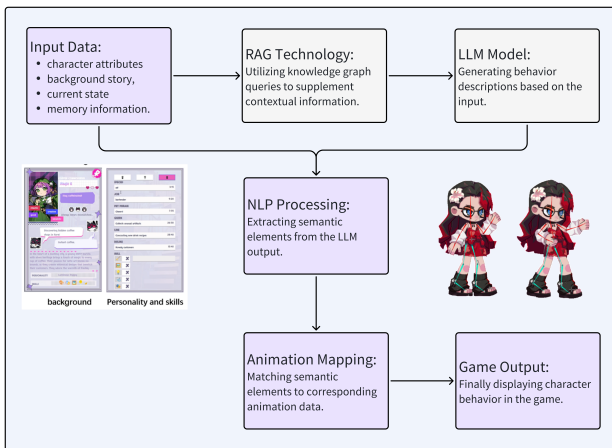


Figure 2: Workflow for In-Game Character Action Driving/Response Based on LLM

3 EVALUATION

3.1 Participants

This study employed a two-phase participant recruitment design. Phase one comprised 12 participants (7 females, 5 males, aged 18-33) recruited through social media platforms. The sample included 3 experienced role-playing game players, 3 professional game developers, and 6 novice users with limited role-playing game experience. Each participant received 30 USD compensation upon study completion. Phase two expanded to 120 participants (55% female, 45% male, aged 18-40). The participant pool consisted of 35% hardcore gamers (>20 hours weekly gameplay), 45% regular gamers (5-20 hours weekly), and 20% casual gamers (<5 hours weekly). All participants committed to a two-week study period, with 50 USD compensation provided upon completion.

All participants were informed of the study’s purpose, procedures, potential risks and benefits, and their rights as research participants. Written informed consent was obtained from all participants prior to their involvement in the study.

3.2 Procedure and Tasks

The entire experimental procedure is illustrated in Figure 3. In Phase One, participants engaged in an in-person session where they spent 20 minutes using our tool to create personalized AI characters, followed by 10 minutes establishing character personalities and backstories. In a sandbox environment, participants interacted with three character types: their created character, a system default character, and a randomly generated character. Semi-structured interviews were conducted post-interaction to discuss the creation process. Participants completed the System Usability Scale (SUS) [1] and Player-Avatar Identification (PAI) [18] questionnaires. The entire first phase, including learning period, creative tasks, interviews, and assessments, lasted approximately 40 minutes.

Phase Two comprised a two-week gameplay period with daily 30-minute interaction sessions. Each session was structured into two components: 15 minutes of mandatory social tasks with designated AI characters, followed by 15 minutes of free exploration and optional character interactions. The data collection process followed two parallel tracks: (1) Bi-daily assessments where participants uploaded relationship evaluation questionnaires and social network formation surveys, and (2) Continuous monitoring through automated interaction logs that captured real-time relationship development and emotional connection metrics. The system automatically recorded all interaction data, character behaviors, and engagement patterns throughout the sessions, while participants submitted structured evaluations every two days to assess relationship dynamics and social network development.

Throughout the study, participants could interact with characters through either text or voice input. We encouraged natural expression and questions, while providing continuous technical support to ensure task completion. This systematic experimental design aimed to examine the formation and evolution of player-AI character relationships.

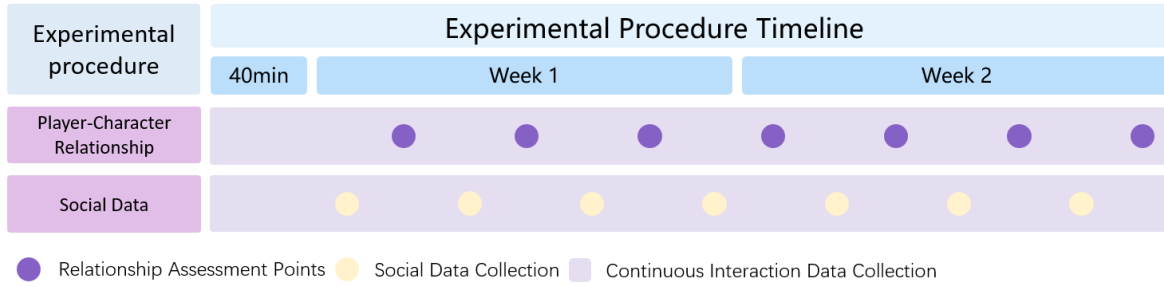


Figure 3: Experimental procedure

4 RESULTS

4.1 Phase 1

All participants successfully completed the design and creation of their game characters within 20 minutes and drove their characters for dialogue and action interactions in the sandbox. Their characters featured a variety of races, hairstyles, and clothing styles (Figure 4), demonstrating that participants could explore imaginative and creative ideas beyond the provided examples, which is quite exciting.



Figure 4: User-generated game characters from the experiment participants: Each game asset consists of a character illustration/tachie (left) and an agent that can be driven in the game engine (right).

The System Usability Scale (SUS) overall score was 82.15 (standard deviation: 9.32), indicating our creation system and sandbox simulation tool are easy to use. For perceived ease of use and learnability, scores were 81.29 (standard deviation: 10.83) and 83.17 (standard deviation: 12.54), respectively. This suggests both novice and experienced role-playing gamers can easily create and drive agent characters and interact with them via natural language text or voice, despite significant differences from traditional parametric creation and mouse-keyboard control methods. This validation

ensured Phase 2 findings wouldn't be compromised by usability issues.

For the Player-Avatar Identification (PAI), participants reported that their created game characters ($M=3.57$, $SD=0.69$) showed statistically significant differences compared to the default generated ($M=3.15$, $SD=0.77$) and randomly generated characters ($M=3.07$, $SD=0.81$). This finding informed our Phase 2 hypothesis that character creation methods substantially influence subsequent player-character relationships.

Beyond quantitative data, our observations and interviews in Phase 1 revealed preliminary relationship patterns between players and their AI characters.

4.2 Phase 2

In the two-week longitudinal study ($N=120$), we systematically observed and analyzed the development of player-AI character relationships. Through a mixed-methods research approach, we collected quantitative data and qualitative feedback to comprehensively understand the evolution of these relationships.

4.2.1 Distribution and Characteristics of Relationship Types. Based on our bi-daily assessments and continuous interaction logs, we applied hierarchical clustering analysis using Ward's method with standardized variables (identification scores, emotional investment ratings, interaction frequency, and relationship quality assessments) to validate the distinct relationship patterns observed in player-character interactions. Analysis of silhouette coefficients across potential clusters (2-cluster: 0.59; 3-cluster: 0.68; 4-cluster: 0.63; 5-cluster: 0.57; 6-cluster: 0.51) statistically confirmed that the three relationship types identified through qualitative observations demonstrated optimal cluster coherence.

The statistical analysis substantiated three significant player-character relationship patterns with distinct characteristics. The self-projection type constituted 37.5% ($n=45$) and exhibited the highest level of character identification ($M=4.2$, $SD=0.65$). Participants tended to view their characters as extensions of themselves, demonstrating a strong need for personalization in character design ($\chi^2(2)=18.43$, $p<.001$). The nurturing type was the most prevalent, accounting for 41.7% ($n=50$). This relationship type demonstrated the strongest emotional investment ($M=4.5$, $SD=0.58$). Participants typically assumed a guardian role, showing sustained attention and responsibility for the character's growth. Repeated measures

ANOVA revealed a significant strengthening trend for this relationship type over the two-week period ($F(1,49)=15.27$, $p<.001$, $\eta^2=0.238$). The companionship type was relatively smaller in proportion (20.8%, $n=25$), exhibiting a moderate degree of identification ($M=3.8$, $SD=0.72$) yet the highest frequency of social interactions. This type of relationship emphasized equality, with participants more inclined to view characters as independent entities.

4.2.2 Qualitative Findings. Regarding perceptions of character autonomy, the majority of participants (82.5%) recognized that characters exhibited unique personalities. One participant noted, "The behavior patterns of the character exceeded my expectations, showcasing distinct personality traits" (P7). In terms of relationship dynamics, over seventy percent of participants (73.3%) reported a gradual improvement in relationship quality. A representative quote from a participant stated, "As the interactions deepened, I came to understand her thoughts and behavior patterns more" (P12). With respect to interaction expectations, participants expressed a strong interest in the expansion of system functionalities, with a particular demand for social features being most prominent (78.3%), followed by character progression systems (71.7%). These findings provide critical insights for the design of social systems for AI characters, particularly in fostering sustained interactions and deep emotional connections. The data demonstrate that participants are capable of forming diverse and meaningful relationships with AI characters, which deepen and develop over time.

5 FEEDBACK AND DISCUSSION

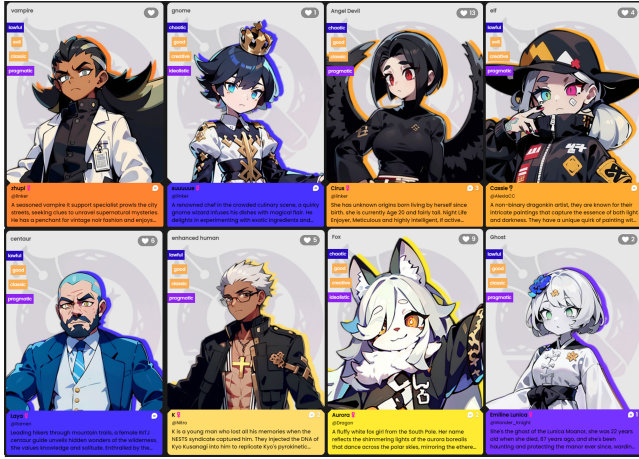


Figure 5: Characters freely created by players in our game community and their descriptions of background and character personality.

5.1 Natural Language-Based Game Character Generation

Through analysis of user feedback, we discovered that the natural language-based character generation method provided players with an entirely novel creative experience. Compared to traditional games' incremental appearance adjustment approaches, this

method was perceived as more innovative and engaging. Players described it as having a "blind box" surprise effect - when their described character suddenly materialized, it often brought unexpected delight. Particularly noteworthy is how this generation method overcomes technical barriers. Even players lacking artistic creation skills can realize their imagined characters through text descriptions. As several participants (P1, P6) expressed, being able to transform their envisioned character images into reality brought immense satisfaction. The system not only accurately captures detailed features such as hairstyles, clothing, and eye colors (P5, P7, P9), but also demonstrates remarkable creative freedom in generating non-conventional character types like elves and vampires.

5.2 Player-Character Relationships

Through longitudinal observation, we discovered diverse relationship patterns between players and AI characters. In the self-projection (myself) type relationship, players view their characters as idealized extensions of themselves. These players tend to invest substantial effort in perfecting appearance and personality details during character creation, seeking high congruence with their own aesthetic preferences and value systems. This suggests game design should provide sufficiently detailed personalization options to enable accurate self-expression.

The nurturing (my child) relationship demonstrates a unique pattern of emotional investment. These players focus primarily on their characters' developmental trajectories, deriving satisfaction from observing the emergence of distinct personalities through interactions. The companionship (my friend) relationship manifests in social interactions, where players regard characters as autonomous entities and seek to establish equitable friendships. They particularly value characters' demonstrations of autonomy and individualized responses in dialogues.

These relationship variations are influenced by multiple factors. Player personality traits play a significant role - introverted players tend to view characters as idealized self-projections, while extroverted players are more inclined to experiment with diverse character types. Age also emerges as a crucial factor, with younger players gravitating toward more imaginative character concepts, while mature players emphasize conversational depth and character development.

Character utility similarly shapes relationship formation. Some players utilize characters as social instruments, prioritizing interactive capabilities within the game, while others view them as vehicles for self-expression, focusing on personalization options.

From a societal impact perspective, AI-generated characters provide a safe "psychological props room" for identity expression. This is particularly significant for marginalized groups, enabling authentic self-expression in virtual spaces. However, this freedom entails responsibility - finding equilibrium between virtual and real worlds, between freedom and regulation, remains an ongoing challenge for future consideration.

6 FUTURE WORK

Future research should extend this work in several key directions. First, building on our findings of distinct player-character relationship patterns, we propose investigating personalized agent

architectures that can dynamically adapt their behavioral strategies and interaction patterns based on identified relationship types (self-projection, nurturing, companionship). This could enhance the emotional resonance and social presence of AI characters. Second, exploring the potential of AI character-based social networks could reveal new paradigms for virtual relationship formation and community building, particularly examining how different relationship patterns manifest in multi-agent environments.

Longitudinal studies are needed to understand the psychological and social impact of sustained interaction with AI characters. This includes examining how these relationships evolve over time and their influence on players' self-perception, emotional wellbeing, and social behaviors. Additionally, investigating the application of AI characters in domains such as education (e.g., language learning, social skills development) and mental health support could reveal valuable insights for designing more effective and empathetic AI companions. These research directions would not only advance our theoretical understanding of human-AI relationships but also inform the development of more sophisticated and socially aware AI character systems.

7 CONCLUSION

This study investigated the impact of AI-generated characters on player identity expression and relationship formation through a natural language-based character creation system in real game environments. Our findings demonstrate that this approach not only enhances the accessibility and creativity of character creation but also transforms players from passive users into active co-creators. Through experimental evaluation, we identified three distinct player-character relationship patterns (self-projection, nurturing, and companionship), reflecting diverse needs for identity expression and emotional connection in virtual spaces. The results suggest that AI-generated characters provide unprecedented freedom for identity exploration while creating safe spaces for self-expression, particularly beneficial for marginalized groups. This research advances our understanding of how AI technology can reshape game character creation and influence player experiences, while also highlighting the importance of recognizing player-AI character relationship dynamics in future game design. These insights provide preliminary exploration for developing more inclusive and emotionally engaging AI character generation and interaction systems.

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